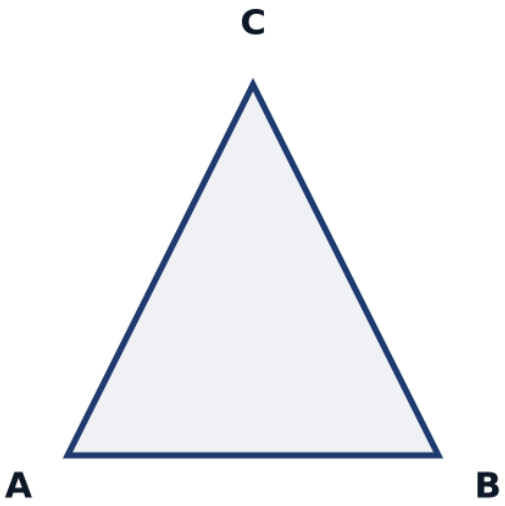
Advanced Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

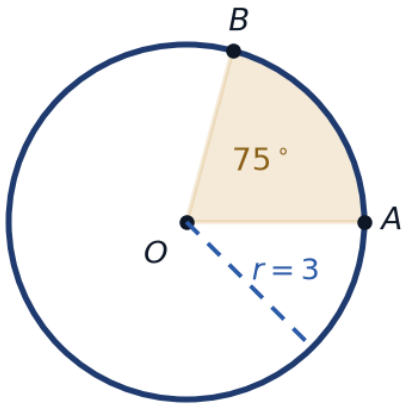
1 [Foundational] • **Target: Distance, midpoint, and slope combined**

Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.



2 [Foundational] • **Target: Equations of circles from given conditions**

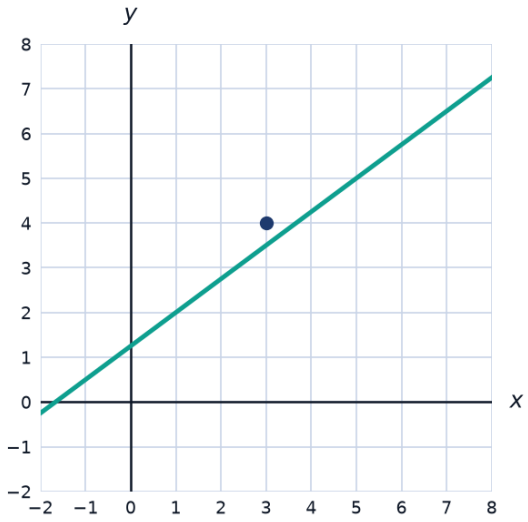
Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.



© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: Distance from a point to a line

Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



4 [Foundational] • Target: Locus problems

Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

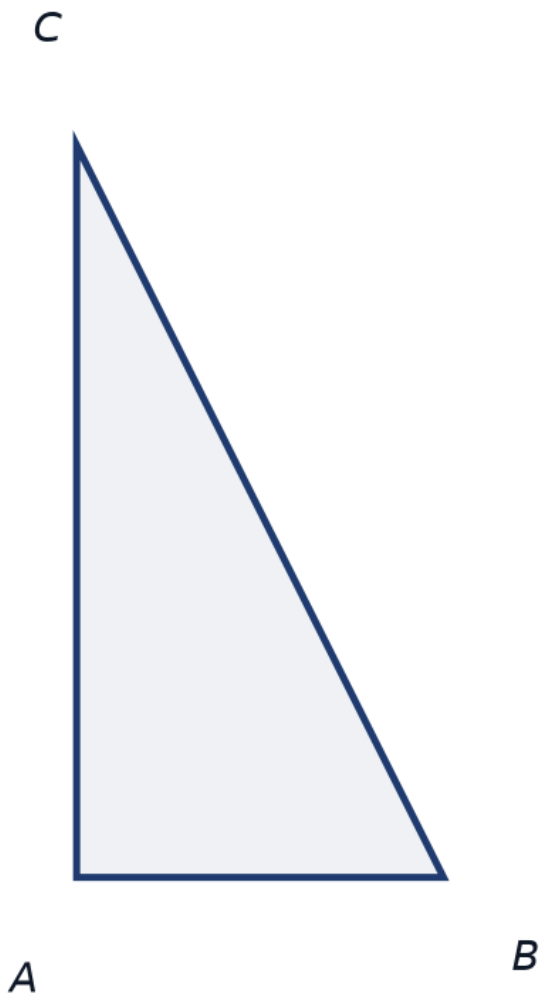
5 [Foundational] • Target: Rotations and general transformations

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Dilations and scale factor

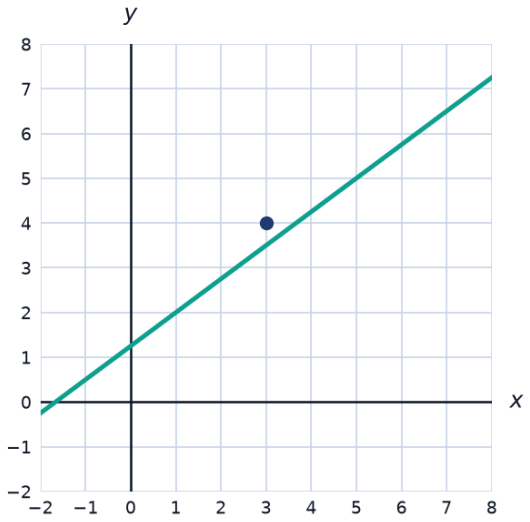
A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.



© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

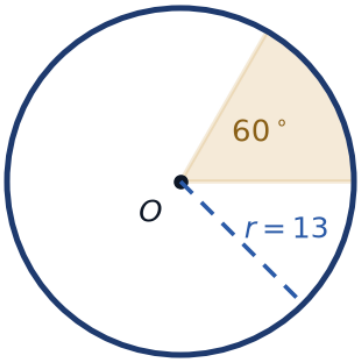
Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A tangent from an external point P touches a circle at T. If the distance from P to the circle's centre O is 13 and the radius is 5, find the length of the tangent segment PT.



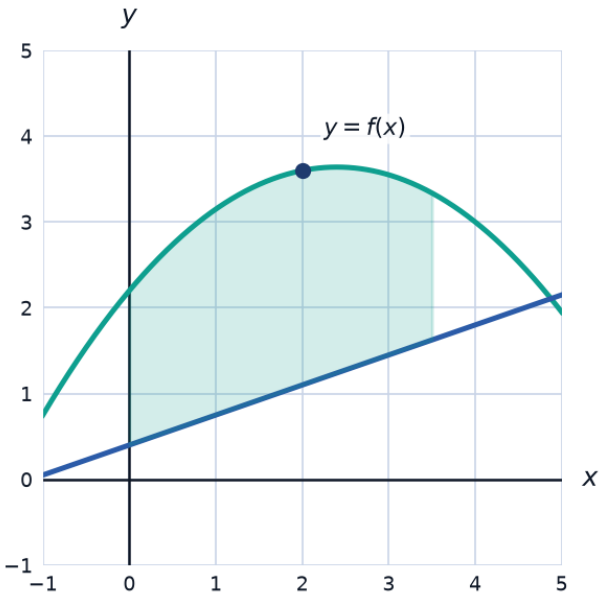
9 [Intermediate] • Target: Finding the equation of a parabola from three points

A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: A multi-part triangle-in-the-plane problem

This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.



11 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

12 [Intermediate] • Target: Translating and reflecting equations of curves

The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.

13 [Intermediate] • Target: Finding vertices of a polygon given transformations

A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

14 [Intermediate] • Target: Finding the intersection of a line and a parabola

Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A multi-part quadrilateral classification problem • Cross-Lesson: 7.01 Advanced Plane Geometry

This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging]** • **Target: Finding the equation of a tangent line to a circle** • **Cross-Lesson: 8.01 Statistics, Counting & Probability**

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

17 **[Challenging]** • **Target: Finding the vertex and axis of symmetry of a parabola from three points** • **Cross-Lesson: 1.01 Advanced Number Theory**

A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 3.01 Advanced Functions**

This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 4.01 Advanced Trigonometry**

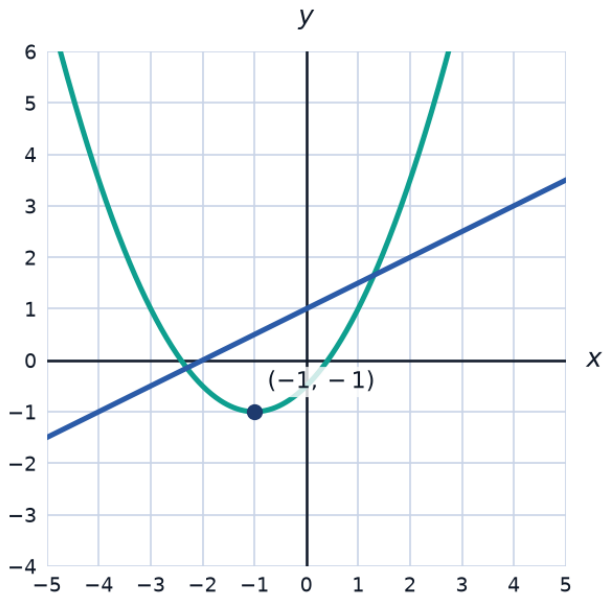
A line passes through $(2, 3)$ and is parallel to $2x - y = 5$. Find its equation in slope-intercept form.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the distance from the point $(9, -1)$ to the line $20x - 21y - 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



a. $\frac{248}{29}$

b. $\frac{219}{29}$

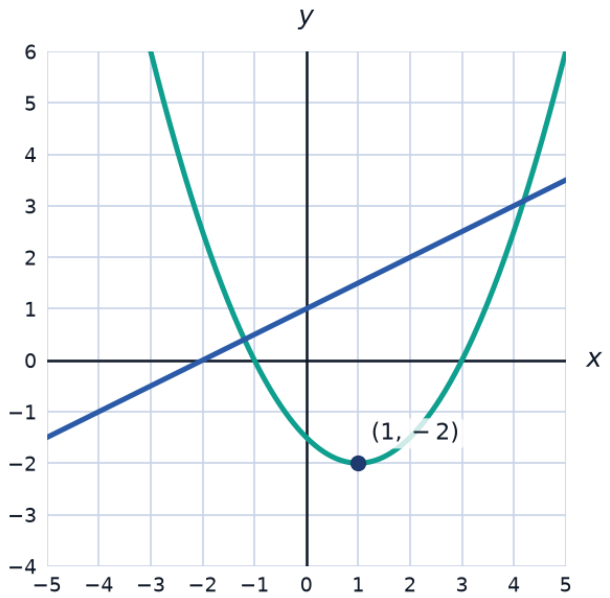
c. -190

d. $\frac{190}{29}$

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the distance from the point $(7, 1)$ to the line $3x + 4y + 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



a. $\frac{44}{5}$

b. $\frac{39}{5}$

c. $\frac{34}{7}$

d. $\frac{34}{5}$

23 [Foundational] • Target: Distance from a point to a line

Find the slope of the line through $(4, -5)$ and $(8, -2)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$

c. $-\frac{4}{3}$

d. $\frac{4}{3}$

24 [Foundational] • Target: Locus problems

Find the slope of the line through $(-6, -5)$ and $(-7, 6)$.

a. -11

b. $-\frac{1}{11}$

c. 11

d. $\frac{1}{11}$

25 [Foundational] • Target: Rotations and general transformations

Find the slope of the line through $(-4, 6)$ and $(5, 9)$.

a. -3

b. $\frac{1}{3}$

c. $-\frac{1}{3}$

d. 3

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Dilations and scale factor

Find the slope of the line through $(-1, -9)$ and $(4, 0)$.

a. $-\frac{9}{5}$

b. $\frac{5}{9}$

c. $\frac{9}{5}$

d. $-\frac{5}{9}$

27 [Foundational] • Target: Finding the equation of a parabola from three points

Find the slope of the line through $(-5, -1)$ and $(-4, 8)$.

a. -9

b. 9

c. $\frac{1}{9}$

d. $-\frac{1}{9}$

28 [Foundational] • Target: A multi-part triangle-in-the-plane problem

Find the slope of the line through $(-3, 7)$ and $(7, -4)$.

a. $-\frac{11}{10}$

b. $\frac{10}{11}$

c. $\frac{11}{10}$

d. $-\frac{10}{11}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

Find the slope of the line through $(-9, -3)$ and $(-7, 1)$.

a. 2

b. -2

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

30 [Intermediate] • Target: Translating and reflecting equations of curves

Find the distance from the point $(1, -1)$ to the line $3x - 4y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{18}{5}$

b. $\frac{13}{5}$

c. $\frac{23}{5}$

d. -13

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Finding vertices of a polygon given transformations

Find the distance from the point $(9, 7)$ to the line $5x + 12y + 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{166}{13}$

b. $\frac{140}{17}$

c. $\frac{140}{13}$

d. $\frac{153}{13}$

32 [Intermediate] • Target: Finding the intersection of a line and a parabola

Find the distance from the point $(-2, -2)$ to the line $7x - 24y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{19}{17}$

b. $\frac{44}{25}$

c. $\frac{69}{25}$

d. $\frac{19}{25}$

33 [Intermediate] • Target: A multi-part quadrilateral classification problem

Find the distance from the point $(-3, 5)$ to the line $12x + 35y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{161}{37}$

b. $\frac{124}{47}$

c. $\frac{198}{37}$

d. $\frac{124}{37}$

34 [Intermediate] • Target: Finding the equation of a tangent line to a circle

Find the distance from the point $(-5, 5)$ to the line $8x + 15y - 13 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{39}{17}$

b. $\frac{22}{23}$

c. $\frac{22}{17}$

d. $\frac{56}{17}$

35 [Intermediate] • Target: Finding the vertex and axis of symmetry of a parabola from three points

Find the distance from the point $(2, -3)$ to the line $20x + 21y + 7 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{16}{41}$

b. $-\frac{16}{41}$

c. $-\frac{16}{29}$

d. $\frac{16}{29}$

36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the y-intercept of the perpendicular bisector of the segment connecting $(-5, -6)$ and $(0, 0)$.

a. $\frac{61}{12}$

b. 0

c. $-\frac{61}{12}$

d. $-\frac{49}{12}$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the y-intercept of the perpendicular bisector of the segment connecting $(2, -3)$ and $(8, 7)$.

a. -5

b. 5

c. 6

d. $-\frac{19}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.

a. -12

b. -4

c. -2

d. -6

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

Find the sum of the x-coordinates of the intersection points of $y = x^2 + 4x - 11$ and $y = -x - 5$.

a. -3

b. -5

c. -6

d. -7

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 8)$, and $(-1, -4)$. Find a.

a. -3

b. 6

c. -2

d. -4

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

A parabola $y = ax^2 + bx + c$ passes through $(0, 7)$, $(1, 0)$, and $(-1, 12)$. Find a.

a. -6

b. -1

c. 0

d. -2

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the distance from the point $(0, 3)$ to the line $8x + 15y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{70}{17}$

b. $\frac{36}{17}$

c. $\frac{53}{17}$

d. $\frac{36}{23}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the distance from the point $(8, 0)$ to the line $12x - 35y - 8 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{162}{37}$

b. $\frac{125}{37}$

c. $\frac{88}{37}$

d. $-\frac{88}{23}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

What are the coordinates of the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$?

a. $(\pm 4, 0)$

b. $(0, \pm 4)$

c. $(\pm 5, 0)$

d. $(0, \pm 3)$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Advanced Algebra & Systems

What is the focal length (distance from center to a focus) of $\frac{x^2}{144} - \frac{y^2}{25} = 1$?

a. 11

b. 12

c. 13

d. 17

© ClipSAT — clipsat.org | All Rights Reserved